

Calvin and Hobbes

by WATTERSON

OH OH BOY OH BOY OH BOY OH BOY OH BOY OH BOY OH BOY OH BOY OH BOY



WAIT? WAIT! I'VE GOT TO SAVOR THIS MOMENT! THE BRILLIANCE OF IT ALL! I'M A GENIUS! A SHEER **GENIUS!**



SUSIE'S PLAYING ON THE SIDEWALK! NOW'S MY CHANCE TO USE THE SNOWBALL I'VE BEEN SAVING IN THE FREEZER?



SHE'LL NEVER EXPECT A SNOWBALL IN **JUNE!** BOL, WILL SHE BE MAD! HA HA HA!



THIS IS GOING TO BE GREAT! HERE IT COMES! OH BOY! OH BOY!



I **MISSED!** DARN IT DARN IT DARN IT!! OF ALL THE MISERABLE LUCK! **AAARRGHH!**



THERE MUST'VE BEEN A CROSS BREEZE! I CAN'T BELIEVE IT! I SAVED THAT SNOWBALL FOR THREE WHOLE MONTHS! I...



I.. I...UH...



THE IRONY OF THIS IS JUST SICKENING.



What does Fine-Tuning result in?

1. Domain Familiarity, domain concepts.
2. I want program "Given X, Give Y"
3. Refusal/Preference Shaping.
4. Preference Alignment (Stop sycophantic behaviour), change personality.
5. Tool Behaviour.

SuperVised Fune Tuning

1. input -> target ouptut
2. prompt -> ideal answer
3. question -> label
4. document -> summary
5. bug-report -> fixed code.

Pre-requisites to use SFT

1. You've very high quality labelled data.
2. Diversified.



How LoRA Works?

Input = $\begin{bmatrix} \end{bmatrix}_{10 \times 100}$
 ↗ words ↘ embedding dim

(DL1) Dense Layer₁ = $\begin{bmatrix} \end{bmatrix}_{100 \times 3000}$

(DL2) Dense Layer₂ = $\begin{bmatrix} \end{bmatrix}_{3000 \times 3000}$
 $\underbrace{\hspace{10em}}_{9 \times 10^6 \text{ parameters}}$

LoRA $\rightarrow$ Low Rank Adapters

Low Rank = 8 (hyper-parameter)

$A = \begin{bmatrix} \end{bmatrix}_{3000 \times 8}$) Randomly Initialized.

$B = \begin{bmatrix} \end{bmatrix}_{8 \times 3000}$
 $24K + 24K = 48K$

DL2 + $A @ B$

$\begin{bmatrix} \end{bmatrix}_{3000 \times 8} \begin{bmatrix} \end{bmatrix}_{8 \times 3000}$
 $\begin{bmatrix} \end{bmatrix}_{3000 \times 3000}$

11 →

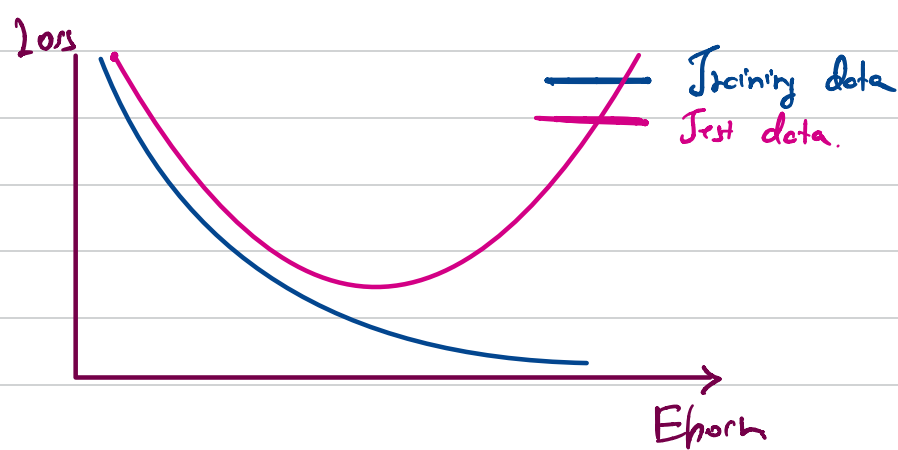
For epoch in 10 epochs

output = fwd-pass(x)

Loss (y, output)

compute gradients

update wts

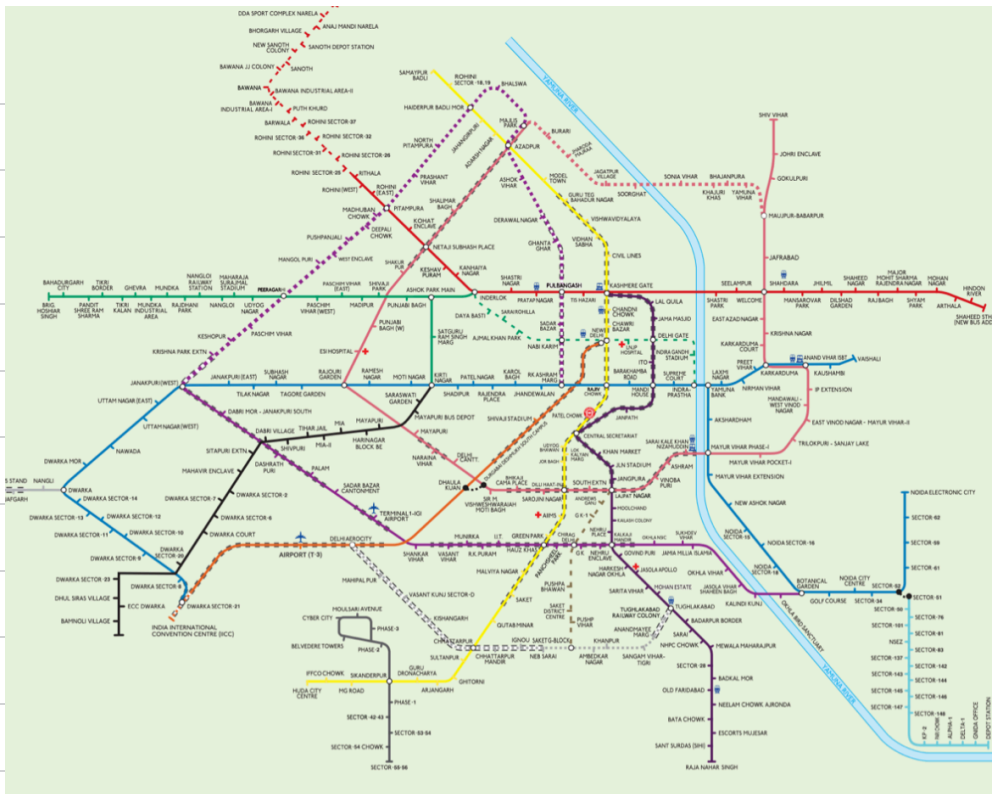


$$W_2 = \begin{bmatrix} \end{bmatrix}_{4512 \times 4512}$$

Low Rank = 16, $A = \begin{bmatrix} \end{bmatrix}_{4512 \times 16}$

$$B = \begin{bmatrix} \end{bmatrix}_{16 \times 4512}$$

$$W_2 = W_2 + \underbrace{\Delta W_2}_{A @ B}$$



https://upload.wikimedia.org/wikipedia/commons/thumb/0/0f/Delhi_metro_rail_network.svg/3840px-Delhi_metro_rail_network.svg.png

Question: Till what age does dystal radius ephysis grow?

Answer: It tends to grow till age of about 21.

Find out Approx memory usage when you load up an LLM like

TinyLlama-1.1B

to train

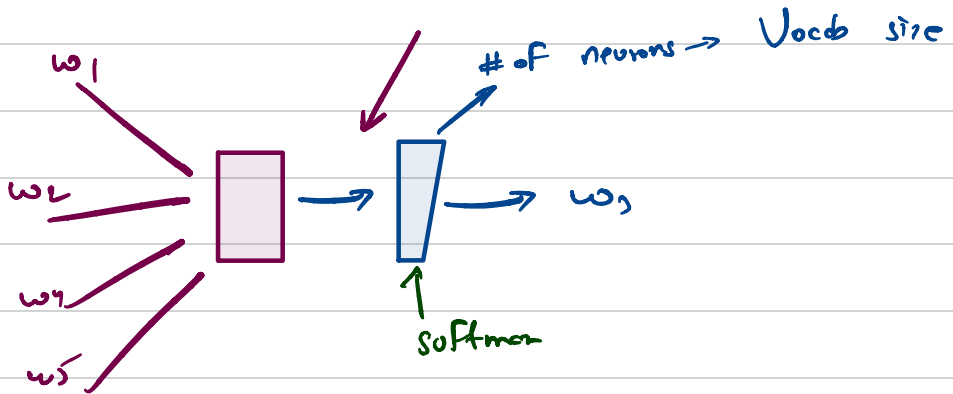
```
messages = [
  {"role": "system", "content": system_prompt},
  {"role": "user", "content": user_message},
]
```

$\langle \text{SYSTEM} \rangle$: " "

$\langle \text{USER} \rangle$: " ' ,

Distillation

A big model $\rightarrow$ acts as a teacher
 A smaller model $\rightarrow$ acts as a student.



What is the color of the apple? $\rightarrow$

Red	0.8
Green	0.19
Blue	0.01

GEN AI Engineer

Checkout requests started failing and API latency has increased suddenly.

1. Restarting some random services.
2. Clearing Caches.
3. Try diff dashboards.
4. Grep logs in 12 different servers, for finding out info on Request failing.
5. Sit in corner and starting chanting mantra.
6. Search slack messages.
7. Apply patch which worked in past.

" When latency is high, restart Service A, B and then G. Then inspect DB, then scale production servers"

Scenario 2:

Senior engineer on day 1:

1. This pattern means DB connection pool Exhaustion.
2. Do not restart anything until you've verified exhaustion.
3. Check these 3 metrics and start service B.

Claude Opus 4.8

Small LLM

Q: What is blue-green deployment

Deepseek: How did you arrive at that solution? help me understand your thought process.

Junior is not exactly same as senior:

1. Senior has still much more deeper world knowledge.
2. Senior has more rare real-world experience.
3. And because senior is a bigger model, it has a more chance of recovery if it comes into a completely unseen situation.